K-ART-4L: Layer-Query Attention over Expert-Annotated Cultural Form Layers for Korean Fine Art Understanding and Generation

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Abstract

Vision–language models inherit their notion of "art" from Western-centric corpora such as WikiArt and the Rijksmuseum collection, leaving the formal vocabulary of Korean fine art—its iconography (도상), technique (기법), materiality (물성), and color symbolism (색채)—structurally invisible to modern multimodal AI. We introduce **K-ART-4L**, an expert-annotated dataset and benchmark that decomposes each artwork into four machine-learnable cultural form layers, annotated by certified curators and art appraisal professionals under a museum-grade licensing and authenticity governance protocol. We propose **Layer-Query Attention** (LQA), a lightweight adapter in which each form layer is represented as a learnable query attending over frozen backbone features, allowing a single encoder to produce disentangled layer-specific representations at negligible additional cost. We further formalize three evaluation constructs: the **Cultural Gap Score** (CGS), which quantifies the retrieval penalty a model pays when queried in the culture's own aesthetic terminology rather than English; the **Layer Fidelity Score** (LFS), a recognizer-as-judge metric verifying that generated images satisfy their layer-wise cultural conditions; and the **Curatorial Turing Protocol** (CTP), a blinded expert evaluation procedure that integrates professional connoisseurship into generative model assessment. Zero-shot diagnosis with CLIP, SigLIP, and DINOv2 on a public-license pilot corpus is designed to expose a systematic cultural representation gap concentrated in the technique and materiality layers. K-ART-4L offers both a resource and a reproducible methodology for extending multimodal AI beyond Western art-historical priors, and a pipeline by which regional museums can convert collections into AI-ready cultural assets.

1 Introduction

Attention-based architectures [1, 3] and large-scale contrastive pretraining [7] have made it possible to align images and language in a shared representation space, and distributed representations have long been known to encode surprisingly precise semantic regularities [2]. Yet what these models learn about art is bounded by what their corpora contain. The de facto standard art corpora—WikiArt [11], the Rijksmuseum challenge set [12], OmniArt [13], and the open-access collections of Western encyclopedic museums—encode a predominantly Western art-historical ontology: oil on canvas, chiaroscuro, impasto, Christian iconography indexed by Iconclass.

Korean fine art does not merely add new classes to this ontology; it requires new axes. The gestural repetition of Park Seo-Bo's *Écriture* (묘법) is not a "style" in the WikiArt sense but a technique defined by embodied action; the absorbency of *hanji* paper and the bleeding of ink (발묵) are material phenomena invisible to canvas-trained texture representations; *obangsaek* (오방색) is not a palette but a directional-symbolic color ontology. When such axes are absent from training data, models silently project Korean art onto the nearest Western category—*Écriture* becomes "minimalist abstract painting," *balmuk* becomes "watercolor bleeding." We call this systematic projection error the **cultural representation gap**.

This paper makes the gap measurable, and proposes a data-and-architecture recipe for closing it. Our contributions are:

C1 (Dataset). K-ART-4L, the first dataset that annotates Korean fine art along four expert-defined cultural form layers—Iconography, Technique, Materiality, Color—with hierarchical labels, bilingual captions, per-image licensing metadata, and a three-tier release structure compatible with Korean public-license (KOGL, 공공누리) and CC0 sources (§3).

C2 (Architecture). Layer-Query Attention (LQA), an adapter that instantiates each form layer as a learnable query vector attending over frozen backbone token features, yielding disentangled layer representations with $< 1\%$ additional parameters (§4.2).

C3 (Evaluation constructs). Formal definitions of the **Cultural Gap Score** for cross-lingual retrieval (§4.4), the **Layer Fidelity Score** for conditional generation (§4.5), and the **Curatorial Turing Protocol**, which converts curatorial connoisseurship into a reliable, blinded evaluation instrument (§4.6).

C4 (Governance pipeline). A reproducible corpus-construction pipeline over Korean public data infrastructure (e-Museum open API, National Heritage Portal, Gongu Madang expired-copyright repository, and CC0 museum APIs), with license-aware routing that physically separates generation-eligible from recognition-only material (§3.3).

2 Background and Related Work

Art datasets and computational art history. Style and genre classification on WikiArt [11] established large-scale art recognition; the Rijksmuseum challenge [12] introduced museum-metadata prediction; OmniArt [13] scaled to over a million artworks. All are overwhelmingly Western in coverage, and none annotates technique or materiality as first-class supervised targets. Iconclass-based iconography tagging presupposes a Western Christian symbol system that has no faithful mapping onto East Asian auspicious iconography (길상 도상). K-ART-4L differs in deriving its taxonomy from Korean art-historical literature and the national heritage standard management schema, and in treating all four form layers as parallel supervised axes.

Cultural bias in vision–language models. Geographic and cultural performance disparities of CLIP-family models [7, 8] are increasingly documented for everyday objects and faces; fine-grained artistic bias —at the level of technique vocabulary and material perception—has lacked a measurement instrument. Our CGS provides one, in the spirit of using a single scalar to summarize a systematic asymmetry.

Attention as an interface to structure. The attention mechanism was introduced to let a decoder softly select relevant source annotations [3], generalized into a self-contained architecture [1]. We reuse this machinery in a compact form: rather than aligning target words to source words, LQA aligns cultural form layers to image tokens, inheriting the interpretability of attention weights—each layer query yields a spatial map showing where in the painting the model looks when judging technique or materiality.

Generative models and cultural heritage. Diffusion-based style learning (LoRA fine-tuning [10] over latent diffusion [9]) makes culturally conditioned generation practical, but evaluation has relied on distribution metrics (FID) and alignment metrics (CLIPScore) that cannot assess cultural validity. A related precedent constructed a Korean traditional painting dataset for a hallyu-focused generative model (Kyung Hee University, Prof. Na Hyung-min's group), but as a closed resource without a public benchmark or standardized evaluation; K-ART-4L is positioned as the open, benchmark-grade counterpart.

Cultural archetype theory (문화원형론). Choi Hye-sil's framework treats cultural archetypes as assets that combine universality with cultural specificity [14]. We operationalize this dichotomy: universality is the portion of an artwork's representation already aligned with the pretrained embedding space; specificity is the residual captured only by layer supervision. The four layers are, in this reading, a factorization of cultural specificity.

3 The K-ART-4L Dataset

3.1 Corpus design

The v1.0 target is 10,000 images in a deliberately small-but-deep regime: annotation density and expert verification, not raw scale, are the differentiators. The pilot corpus (1,000 images) used for all diagnosis experiments in this paper is drawn exclusively from public-license sources, organized into six blocks (Table 1).

Table 1: Composition of the 1,000-image public-license pilot corpus.

Block	Genre	Size	Primary source	Layer emphasis
A	Minhwa (작호도·책가도·십장생·문자도)	300	e-Museum API; Gongu Madang	Iconography
B	Genre painting (풍속화)	200	e-Museum; CC0 museum APIs	Iconography, Color
C	Buddhist painting (탱화·괘불)	150	National Heritage Portal	Color, Iconography *(recognition-only)*
D	Court & decorative art (일월오봉도·단청)	150	Heritage Portal; Nat'l Palace Museum	Color, Pattern
E	Modern ink painting, expired copyright	150	Gongu Madang (만료저작물)	Technique *(generation-eligible)*
F	Materiality reference crops	50	NMK high-resolution service	Materiality
Total		**1,000**		

Two routing rules are enforced at the pipeline level: (i) generative experiments draw only from Block E (copyright-expired works), and (ii) religious art (Block C) is flagged recognition-only, reflecting cultural-appropriation governance. Foreign CC0 holdings (10–15% of Blocks A–B) are intentionally mixed in so that digitization-quality disparities between domestic and international collections become an analyzable variable rather than a confound.

3.2 Four-layer annotation schema

Each image x receives labels $y = (y_{\text{icon}}, y_{\text{tech}}, y_{\text{mat}}, y_{\text{color}})$ and a bilingual free-text caption.

L1 Iconography (도상). Three-level hierarchy—category (auspicious / religious / everyday / pattern) $\rightarrow$ motif ($\sim$120 items: tiger, magpie, peony, …) $\rightarrow$ symbolic meaning (protection 벽사, wealth 부귀, longevity 장수). A crosswalk table to Iconclass is released for interoperability.

L2 Technique (기법). $\sim$40 items covering brush methods (구륵/몰골), ink behaviors (발묵/파묵), repetition-based gestural techniques (묘법류), and back-coloring (배채법); each item carries an action description usable directly as a generation prompt fragment.

L3 Materiality (물성). Support (*hanji* variants, silk, canvas), pigment family, and surface behavior (absorption, bleeding, relief); an RTI-imaged subset ships with normal maps.

L4 Color (색채). *Obangsaek*/*ogansaek* ontology labels, a perceptual palette (k-means in CIELAB, $k = 8$), and the symbolic color–direction graph.

Annotation follows a two-stage metadata-bootstrapped protocol: source metadata fields (재질, 주제분류) are rule-mapped to seed labels, then verified and refined by a panel of certified curators (학예사) and appraisal professionals; inter-annotator agreement is reported per layer as Krippendorff's α with hierarchical distance weighting. The full 30-page codebook is released with the dataset.

3.3 Licensing and release

A three-tier structure—(a) public-domain / KOGL Type-1, redistributable; (b) institution-licensed, distributed as URL + metadata + annotations (LAION-style reference release); (c) restricted, metadata-only behind an API gate—is recorded per image, since KOGL type varies within a single institution. KOGL

Types 3–4 (no-derivatives) are physically quarantined from every generative pipeline. All digitized files carry C2PA content credentials, and every generated output in the benchmark is required to embed AI-provenance marking.

4 Method

4.1 Problem formulation

Let f_θ be a frozen pretrained encoder mapping an image x to token features $H \in \mathbb{R}^{n \times d}$ (n patch tokens, width d). K-ART-4L defines three tasks:

T1 Multi-layer attribute recognition, predicting the label tuple y.

T2 Cross-lingual art retrieval between Korean aesthetic-terminology queries and images.

T3 Layer-conditioned generation, sampling an image $G(c)$ from a condition tuple $c = (c_{\text{icon}}, c_{\text{tech}}, c_{\text{mat}}, c_{\text{color}})$.

4.2 Layer-Query Attention (LQA)

We instantiate each form layer $\ell \in \{\text{icon, tech, mat, color}\}$ as a learnable query vector $\mathbf{q}_\ell \in \mathbb{R}^d$ and compute the layer representation by scaled dot-product attention [1] over the frozen tokens:

$$\mathbf{z}_\ell \;=\; \text{Attention}\big(\mathbf{q}_\ell W_\ell^Q,\; HW_\ell^K,\; HW_\ell^V\big) \;=\; \text{softmax}\left(\frac{\mathbf{q}_\ell W_\ell^Q \big(HW_\ell^K\big)^\top}{\sqrt{d_k}}\right) HW_\ell^V \tag{1}$$

with layer-specific projections $W_\ell^Q, W_\ell^K \in \mathbb{R}^{d \times d_k}$, $W_\ell^V \in \mathbb{R}^{d \times d_v}$, extended to h heads exactly as in multi-head attention [1]. A linear classifier head g_ℓ maps $\mathbf{z}_\ell$ to layer-ℓ logits. Because the backbone is frozen and each layer adds only one query and three projections, the adapter contributes $< 1\%$ of backbone parameters, and—mirroring the interpretability argument of attention models [3]—the attention distribution in (1) is a *spatial evidence map*: the pixels the model consults when judging, e.g., materiality. We report these maps as a qualitative diagnostic (which regions "look like *hanji*" to the model).

Design rationale (why layer queries rather than four separate heads on pooled features). A single pooled `[CLS]` feature entangles all layers; separate pooled heads share one spatial summary and cannot specialize. Layer queries let each axis attend to its own evidence—iconography to figural regions, materiality to surface texture—at $O(n \cdot d)$ per layer, the same asymptotic cost as one additional attention row.

4.3 Training objective

Layers differ in label cardinality and imbalance, so per-layer losses are class-balanced focal losses $\mathcal{L}_\ell$ [15, 16], combined with homoscedastic uncertainty weighting [17]:

$$\mathcal{L} \;=\; \sum_\ell \left[\frac{1}{2\sigma_\ell^2}\,\mathcal{L}_\ell \;+\; \log \sigma_\ell\right] \tag{2}$$

where σ_ℓ are learned. For the RTI subset, an auxiliary normal-map regression loss is added to the materiality branch. Hierarchical labels in L1 use a hierarchy-aware target distribution so that a motif-level error within the correct category is penalized less than a category-level error; we report hierarchical F1 (hF1) alongside macro-F1 and mAP.

4.4 Cultural Gap Score (CGS)

For T2, let $R@k(q; L)$ denote Recall@k for query set q phrased in language L over the same image pool. We define

$$\text{CGS@}k \;=\; R@k(\text{en}) \;-\; R@k(\text{ko}) \tag{3}$$

computed over paired queries that are professional translations of one another (e.g., "landscape with pronounced ink bleeding" $\leftrightarrow$ "발묵의 번짐이 두드러진 산수"). CGS@$k > 0$ quantifies the penalty a model

pays when addressed in the culture's own critical vocabulary; CGS ≈ 0 for a model that has genuinely internalized that vocabulary. We report CGS per layer by restricting query sets to layer-specific terminology, which localizes where in the form-layer space the representation gap resides.

4.5 Layer Fidelity Score (LFS)

For T3, distribution metrics (FID) cannot verify that a generated image satisfies its cultural conditions. Let $\hat{f}_\ell$ be the T1 recognizer for layer ℓ (the *judge*), G a conditional generator, and $\{c^{(i)}\}_{i=1}^{N}$ sampled condition tuples. We define

$$\mathrm{LFS}_\ell \;=\; \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\Big[\,\hat{f}_\ell\big(G(c^{(i)})\big) = c_\ell^{(i)}\,\Big], \qquad \mathrm{LFS} \;=\; \frac{1}{4}\sum_{\ell} \mathrm{LFS}_\ell \tag{4}$$

i.e., the rate at which the judge re-recognizes each conditioned layer in the output. LFS is reported alongside FID and CLIPScore; the three metrics are complementary (realism, text alignment, cultural-condition satisfaction). To guard against judge–generator co-adaptation, the judge is trained on a disjoint split and frozen before any generative experiment.

4.6 Curatorial Turing Protocol (CTP)

CTP is a blinded expert procedure with three components.

(i) A panel of $m \geq 7$ certified curators and appraisers rates mixed sets of authentic and generated images on four 7-point scales—technique plausibility, materiality rendition, iconographic coherence, cultural dissonance—with inter-rater reliability reported as ICC.

(ii) A discrimination task measures panel accuracy at separating authentic from generated works; accuracy approaching chance (0.5) is treated as a measurement of material-technical rendition quality, *never* as a deployment goal—all distributed outputs carry mandatory AI-provenance marking (§3.3).

(iii) Automatic-metric validation: Spearman correlation between LFS and panel ratings determines whether LFS can serve as a low-cost proxy for expert judgment, which—if confirmed—is itself a reusable methodological result for cultural-heritage generation research.

5 Experiments

Integrity note. All result cells below are placeholders pending the pilot run specified in the released corpus-construction plan; hypotheses H1–H3 are pre-registered here so that the pilot constitutes a confirmatory, not exploratory, analysis. Pilot scale (1,000 public-license images, single-GPU training) is chosen so that every experiment is reproducible on one A100.

5.1 Setup

Backbones: CLIP ViT-L/14 [7], SigLIP-So400m [8], DINOv2-L [18]. **Regimes:** frozen zero-shot, LQA adapter, LoRA [10], full fine-tune. **Generation:** SDXL [19] with per-layer LoRA ensembles; color conditions injected as palette hints. Optimizer AdamW with cosine schedule; splits are stratified by artist and period with artist-disjoint partitions (70/10/20) to prevent leakage.

5.2 T1: multi-layer recognition

Pre-registered hypotheses.

H1 Zero-shot performance degrades in the order iconography $>$ color $>$ technique $>$ materiality, because technique and materiality are rarely verbalized in web alt-text.

H2 CGS is maximal for the technique layer—the layer whose vocabulary is least translated.

H3 In generation, $\mathrm{LFS}_{\mathrm{color}}$ is highest and $\mathrm{LFS}_{\mathrm{mat}}$ is the bottleneck, indicating that texture-aware generation is the open problem this benchmark should drive.

Table 2: Macro-F1 (%) by layer — placeholder.

Model	Icon.	Tech.	Mat.	Color	Avg	hF1
CLIP zero-shot	[xx]	[xx]	[xx]	[xx]	[xx]	[xx]
SigLIP zero-shot	[xx]	[xx]	[xx]	[xx]	[xx]	[xx]
DINOv2 + linear probe	[xx]	[xx]	[xx]	[xx]	[xx]	[xx]
DINOv2 + LQA (**ours**)	[xx]	[xx]	[xx]	[xx]	[xx]	[xx]
+ RTI auxiliary loss	—	—	[xx]	—	—	—

5.3 T2: cross-lingual retrieval

We report R@$\{1, 5, 10\}$, MRR, and per-layer CGS@10 (Eq. 3) for each backbone, before and after LQA adaptation. The diagnostic claim is falsifiable: if CGS ≈ 0 already at zero shot, the cultural representation gap thesis is refuted for retrieval.

5.4 T3: layer-conditioned generation

Generators are trained only on Block E (expired copyright). We report FID, CLIPScore, LFS (Eq. 4), and full CTP results including ICC and the LFS–panel Spearman ρ. Qualitative panels contrast condition tuples that differ in exactly one layer (e.g., same iconography, 몰골 vs. 구륵 technique), the layer-conditioned analogue of controlled generation probes.

5.5 Ablations

(i) LQA vs. shared-pooling multi-head baseline (does per-layer attention matter?).
(ii) Uncertainty weighting (Eq. 2) vs. uniform λ.
(iii) Metadata-bootstrapped seeding vs. from-scratch annotation, measured as expert correction rate and wall-clock annotation cost.
(iv) With vs. without foreign CC0 mixing, isolating digitization-quality effects.

6 Discussion, Limitations, and Broader Impact

Applications. The T2 model transfers directly to museum collection search, AI docent systems, and visitor-personalized curation; the pipeline is an implementable reference for digitally native museum design—institutions built after this benchmark (e.g., a provincial museum of art scheduled to open in 2029 in Gyeongsangbuk-do) can embed the archive–annotation–benchmark loop at the design stage rather than retrofitting it.

Limitations.

(i) v1.0 scale suits fine-tuning and evaluation, not pretraining.
(ii) Iconographic symbolism admits scholarly disagreement; v2 will release multi-annotator label distributions rather than forced single labels.
(iii) CTP panels are Korea-based; cross-cultural panel replication is future work.
(iv) The cultural-appropriation risk of generative reinterpretation is mitigated by governance (recognition-only flags, provenance marking) but not eliminated; we state this explicitly rather than claiming a technical fix.

Broader impact. CGS generalizes to any culture whose critical vocabulary is under-represented in web text; we intend K-ART-4L's schema, codebook, and governance pipeline to serve as a template for parallel efforts in other non-Western art traditions.

7 Conclusion

We presented **K-ART-4L**, a four-layer expert-annotated dataset and benchmark for Korean fine art; **Layer-Query Attention**, a near-free adapter that disentangles cultural form layers over frozen backbones; and three evaluation constructs—CGS, LFS, CTP—that make cultural representation gaps measurable in recognition, retrieval, and generation respectively. The broader claim is methodological: closing cultural gaps in multimodal AI is not primarily a scaling problem but a *supervision-structure* problem, and the unit of supervision should be the culture's own formal categories, defined and verified by its own experts.

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Appendix A Reproducibility checklist (NeurIPS-style)

- Annotation codebook (30 pp.) and raw inter-annotator agreement data released with the dataset.
- Three-tier license structure documented per image; KOGL Types 3–4 quarantined from generation; C2PA credentials on all files.
- Training code, hyperparameters, seeds released; pilot reproducible on a single A100.
- Datasheets for Datasets [20] and Model Cards provided.
- Pre-registered hypotheses H1–H3 (§5.2) fixed before pilot execution.

Appendix B 투고 노트 (저자용 — 투고 시 삭제)

편집 주의. 이 부록은 저자 내부용이며 최종 투고본에서 반드시 삭제한다.

이 v2.0은 v1.0(국문 초안) 대비 다음이 바뀌었다.

① 전면 영문화 및 NeurIPS/ICLR 형식 규범 적용(번호 수식, 번호 인용, 비교표).

② 산문으로만 있던 평가 개념을 수식 (3)(4)로 형식화 — Bahdanau et al.이 '정렬'을 수식으로 형식화해 기여로 만든 방식과 동형.

③ 아키텍처 기여(LQA, 수식 (1))를 신설해 데이터셋-only 논문에서 데이터셋+방법 논문으로 격상 — Attention Is All You Need의 §4 "Why Self-Attention"을 본떠 설계 정당화 문단 포함.

④ 가설 사전등록(H1–H3)으로 파일럿을 확증적 분석으로 전환.

워크숍(4p) 투고 시에는 §4.6과 어블레이션을 축약하고 T1·T2 진단 결과만으로 구성한다. KCC 국문판은 본문 번역 + 국내 사례(문화원형론) 확장으로 별도 조판한다.